

**Amrita School of Engineering,
Bangalore-35**

23MAT117-Linear Algebra

Lab Session Sheet -8

Projections onto subspaces, Least Square Approximation, Linear Regression

- Projection of a vector $\mathbf{v}$ on a subspace S , which is a column space of matrix A (with all independent columns) is $\mathbf{p} = A(A^T A)^{-1} A^T \mathbf{v}$.

If some columns in A are dependent (i.e., if $\text{Rank}(A_{m \times n}) < n$), then the projection of a vector $\mathbf{v}$ on a subspace S , which is a column space of matrix A is $\mathbf{p} = A * \text{pinv}(A) * \mathbf{v}$, where $\text{pinv}(A)$ is the pseudo inverse of A .

1. Find the projection of $\mathbf{v} = \begin{pmatrix} -3 \\ -3 \\ 8 \\ 9 \end{pmatrix}$ on the space spanned by $\begin{pmatrix} 3 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ -1 \end{pmatrix}$.
 2. Find the projection of $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ on the space spanned by $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$.
 3. Find the projection of $\mathbf{u} = \begin{pmatrix} a \\ b \\ 9 \end{pmatrix}$ on the column space of $A = \begin{pmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 3 & 6 \\ 0 & 1 & 1 & 2 \end{pmatrix}$ where a is the last two digits of your registration number and b is your date of birth.
- Special case: If $\mathbf{A}$ is a column vector $\mathbf{a}$, i.e., Projection of a vector $\mathbf{v}$ onto a line with basis $\mathbf{a}$ is $\mathbf{p} = \mathbf{a} \frac{\mathbf{a}^T \mathbf{v}}{\mathbf{a}^T \mathbf{a}}$.
4. Find the projection of $\mathbf{v} = \begin{pmatrix} 1 \\ -2 \\ 8 \\ 1 \end{pmatrix}$ on the line with parametric form, $\mathbf{r}(t) = \begin{pmatrix} t \\ 2t \\ 3t \\ -t \end{pmatrix}$
 5. Find the projection of $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \\ 7 \end{pmatrix}$ on the line of intersection of the planes, $2x - y + z = 0$ and $x - y + 3z = 0$.
 6. Find the projection of $\mathbf{v} = \begin{pmatrix} 1 \\ a \\ b \end{pmatrix}$ on the orthogonal complement of the plane, $x + 2y + 3z = 0$ if a is the last two digits of your registration number and b is your date of birth.

- Least square solution of a linear system, $Ax = b$ is:
 $\hat{x} = (A^T A)^{-1} A^T b$, if all the column vectors in A are independent (i.e. if $A^T A$ is invertible).
 $\hat{x} = \text{pinv}(A) * b$, if the column vectors in A are not all independent.

Fitting of a Regression line using excel:

	A	B	C	D	E	F	G	H	I	J	K
1		x	y	x _i y _i	x _i ²						
2		-10	-130.771	1307.711	100						
3		-9	-124.911	1124.2	81						
4		-8	-188.656	1509.25	64						
5		-7	-87.4707	612.2951	49			n=21			
6		-6	3.996329	-23.978	36						
7		-5	-133.754	668.7695	25			B Matrix			b vector
8		-4	-69.7174	278.8696	16						
9		-3	21.11265	-63.338	9			770	0		17078.85
10		-2	5.479887	-10.9598	4			0	21		1148.982
11		-1	0.237035	-0.23704	1						
12		0	41.39672	0	0						
13		1	42.67804	42.67804	1						
14		2	155.2094	310.4189	4						
15		3	109.763	329.289	9						
16		4	172.5026	690.0104	16						
17		5	155.0395	775.1975	25						
18		6	203.447	1220.682	36						
19		7	229.9432	1609.603	49						
20		8	238.0019	1904.016	64						
21		9	260.1743	2341.569	81						
22		10	245.2808	2452.808	100						
23		Sums	0	1148.982	17078.85	770					

Since $\sum x_i = 0$ in our Example

$$m = \frac{\sum x_i y_i}{\sum x_i^2}$$

m= 22.18033
c= 54.71343

$$c = \frac{\sum y_i}{n}$$

Regression using MATLAB

	A	B	C
1	x	y	
2	-10	-130.771	
3	-9	-124.911	
4	-8	-188.656	
5	-7	-87.4707	
6	-6	3.996329	
7	-5	-133.754	
8	-4	-69.7174	
9	-3	21.11265	
10	-2	5.479887	
11	-1	0.237035	
12	0	41.39672	
13	1	42.67804	
14	2	155.2094	
15	3	109.763	
16	4	172.5026	
17	5	155.0395	
18	6	203.447	
19	7	229.9432	
20	8	238.0019	
21	9	260.1743	
22	10	245.2808	

Write a MATLAB code to find the regression line and also plot it along with the given data points

- Create an excel file with your data as in the figure. Name it as xlformatlabReg.xls and use the code below:

```
[B,text]=xlsread('xlformatlabReg.xlsx');
x=B(:,1);
y=B(:,2);
n=length(x);
A=[x,ones(n,1)];
ATA=[sum(x.^2),sum(x);sum(x),n];
ATb=[sum(x.*y);sum(y)];
m_and_c=inv(ATA)*ATb;
y_projected=A*inv(ATA)*ATb;
plot(x,y,'*');
hold on;
plot(x,y_projected);
```

Linear Regression : Generation of data and fitting

Generate a data with $m=7$, $c=3$ and a noise standard deviation 6. With this data perform regression and verify the m and c values. Also plot the generated points and the fitted line

```
m=7;
c=3;
x=-5:0.1:5; % a row vector x
n=length(x);
y=m*x+c;
noise_stddev=6; % noise standard deviation

noise=noise_stddev*randn(1,n);
yn=y+noise;
plot(x,yn,'*'); % plot of generated points

hold on % fitting a line or re-estimating m and c
% create matrix A with x and ones

A=[x' ones(n,1)];
yn=yn'; % make yn a column vector
m_and_c=pinv(A)*yn; % estimate m and c
y_projected=A*pinv(A)*yn; % y estimated
plot(x,y_projected);
m_and_c
```

Estimation of m and c

Step 1: Generate the data for regression with noise

```
m = 9;
c = 5;
x = (-25:25)';
y_true = m * x + c;
noise_std = 3;
noise = noise_std * randn(size(x));
y = y_true + noise;
```

% Step 2: Perform linear regression to estimate m and c

```
A = [x, ones(size(x))];
```

```

params = A\y; % Parameters estimation using least squares
m_estimated = params(1);
c_estimated = params(2);

% Step 3: Plot the original data and regression line
figure;
plot(x, y, 'bo', 'MarkerSize', 5); % Original data points
hold on;
plot(x, m_estimated*x + c_estimated, 'r-', 'LineWidth', 2); % Regression line
hold off;
xlabel('x');
ylabel('y');
title('Linear Regression');
legend('Data with Noise', 'Regression Line');

```

8. Consider the following data:

xi	yi
-5	-16.5
-4.5	-15.2
-4	-14.9
-3.5	-13.3
-3	-10.3
-2.5	-8.9
-2	-6.8
-1.5	-7.8
-1	-5.9
-0.5	-4.3
0	-1.3
0.5	0.1
1	2.2
1.5	1.2
2	3.1
2.5	4.7
3	7.7
3.5	9.1
4	11.2
4.5	10.2
5	15.2

- Find the equation of a regression line for the following data using excel.
- Enter the data in an excel sheet, and then find the regression line using MATLAB. Also plot the regression line and the data points in the same figure.

Practice questions:

1. Generate a data for regression with a fixed $m=9$, $c=5$ for $x = -25, -24, -23, \dots, 24, 25$ and introducing a noise with standard deviation 3.
2. Using the data generated above, re-estimate the values of m and c . Also find the regression line and plot it in MATLAB.
3. Generate a data for regression with a fixed $m=-7$, $c=2$ for $x = 1, 2, \dots, 15$ and introducing a noise with standard deviation 5.
4. Using the data generated above, re-estimate the values of m and c . Also find the regression line and plot it in MATLAB.
5. Find the least square solution for the following systems:
 - a. $x + y = 10$; $2x + y = 10$; $3x + y = 16$; $4x + y = 27$; $5x + y = 23$.
 - b. $2x - 3y + z = 1$; $x + 3y - 4z = -4$; $3x - 2y - z = -1$; $5x - y - 4z = -4$.
 - c. $2x + y = 20$; $2x - 3y = 4$; $3x + y = 28$; $4x + y = 36$;
 - d. $x + 3y - 5z = -1$; $2x - 3y + 8z = 7$; $4x + 3y - 2z = 5$; $9x - 3y + z = 8$; $3x - 5y + 3z = 0$.